Structured notes: Section 3: emf of a rotating coil; mean, r.m.s. and virtual values (Murugeshan 13.1)

Murugeshan, *Electricity and Magnetism* (10th ed.), Ch. 12 §12.1 and §12.2 (Module 4, syllabus item 15–16). Figures are drawn for these notes from the book formulas; the book figure number is in each caption. Problems are not included.

13.1 EMF induced in a rotating coil

Source: book-sources/m4-s3-ac-emf.html, Murugeshan pp. 329–333 (printed). Syllabus item 17. The book does not number the equations of this section; they are shown unnumbered.

Prerequisites

- Magnetic flux and Faraday's law (pre-physics P4).
- Sinusoidal functions and their integrals (pre-math M5, M7).

1. Set-up

A rectangular coil of N turns, sides a and b, rotates at angular velocity ω about an axis perpendicular to a uniform field $\mathbf{B}$ (Fig. 13.1). The normal makes angle $\theta = \omega t$ with $\mathbf{B}$ at time t, taking $\theta = 0$ at the instant the coil plane is perpendicular to $\mathbf{B}$.

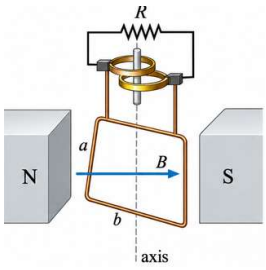


Fig. 13.1 (book): rectangular coil rotating between poles N and S, with slip rings and resistance R.

$$\phi = NBA\cos\omega t, \qquad A = ab$$

2. Induced emf

$$E = -\frac{d\phi}{dt} = E_0 \sin\omega t, \qquad E_0 = NBA\omega$$

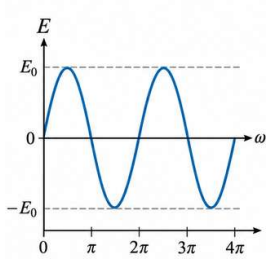


Fig. 13.2 (book): emf $E = E_0 \sin \omega t$ against ωt; a sine curve.

E_0 is the peak value. The period is $T = 1/\nu$ and $\omega = 2\pi\nu$ (Fig. 13.3).

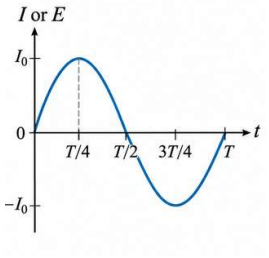
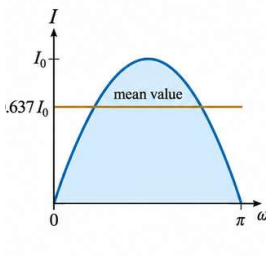


Fig. 13.3 (book): one cycle of ac voltage or current, with peak value and the times T/4, T/2, 3T/4, T.

3. Mean, r.m.s. and virtual values

$$I_{mean} = \frac{2I_0}{\pi} = 0.637\, I_0, \qquad E_{mean} = 0.637\, E_0$$



Mean value over half a cycle, 0.637 I_0 (book, §13.1, shaded area).

$$\overline{I^2} = \frac{I_0^2}{2}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}} = 0.707\, I_0$$

$$E_{rms} = \frac{E_0}{\sqrt{2}}$$

The r.m.s. value equals the virtual (effective) value: the direct current giving the same heat in the same resistance.

4. Form factor and half-cycle mean

$$\begin{aligned}\text{Form factor} &= \frac{I_{rms}}{I_{mean}} \\ &= \frac{0.707}{0.637} \\ &= 1.11\end{aligned}$$

Over half a period, $E_{av} = 2E_0/\pi$.

5. Impedance and reactance

The impedance is $Z = E/I$; reactance is the part due to L or C (X_L, X_C); admittance is $Y = 1/Z$.